ASS 17.5

AI ASSISTED CODING

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```
# Import libraries
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, LabelEncoder

# Load dataset
data = pd.read_csv("loanapproval.csv")

# Preview data
print("Initial Data:")
print(data.head())
print("\nMissing Values:\n", data.isnull().sum())


# -----------------------------
# 1. Handle Missing Values
# -----------------------------
# Fill numerical columns with median (robust choice)
data['loan_amount'].fillna(data['loan_amount'].median(), inplace=True)
data['credit_score'].fillna(data['credit_score'].median(), inplace=True)


# -----------------------------
# 2. Encode Target Variable
# -----------------------------
# Convert loan_status (Approved/Rejected) → (1/0)
le = LabelEncoder()
data['loan_status'] = le.fit_transform(data['loan_status'])
```

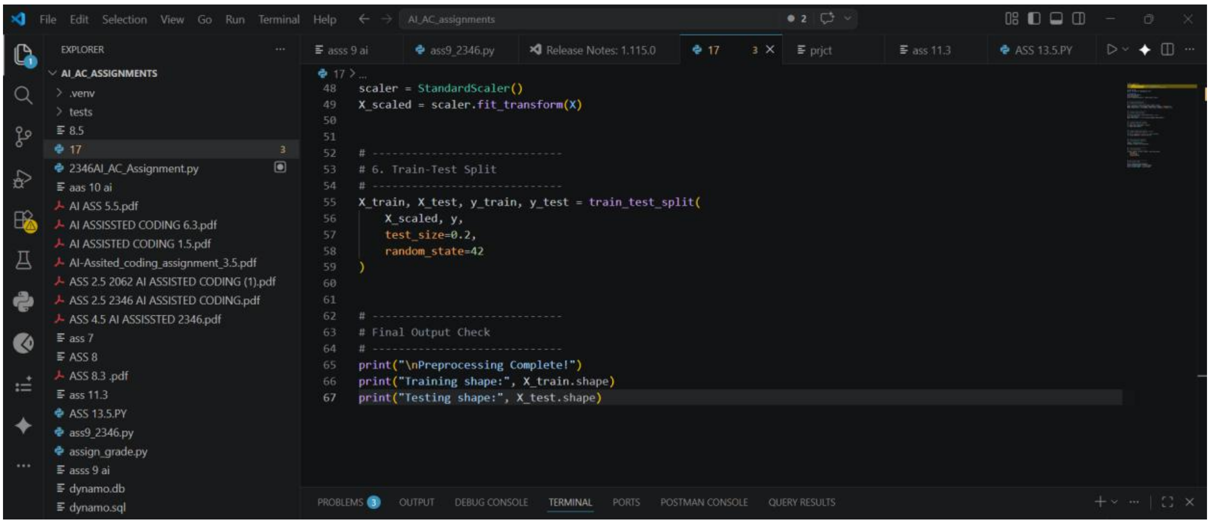
```python
# -----------------------------
# 3. Separate Features & Target
# -----------------------------
X = data.drop('loan_status', axis=1)
y = data['loan_status']


# -----------------------------
# 4. Handle Categorical Features (if any)
# -----------------------------
# Convert categorical columns into dummy variables
X = pd.get_dummies(X, drop_first=True)


# -----------------------------
# 5. Scale Numerical Features
# -----------------------------
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)


# -----------------------------
# 6. Train-Test Split
# -----------------------------
X_train, X_test, y_train, y_test = train_test_split(
    X_scaled, y,
    test_size=0.2,
    random_state=42
)
```

```
# -----------------------------
# Final Output Check
# -----------------------------
print("\nPreprocessing Complete!")
print("Training shape:", X_train.shape)
print("Testing shape:", X_test.shape)
```



Social Media Profiles

Preprocess a social media engagement dataset using AI

assistance.

Instructions:

• Remove duplicate user profiles.

• Handle missing values

• Encode categorical columns

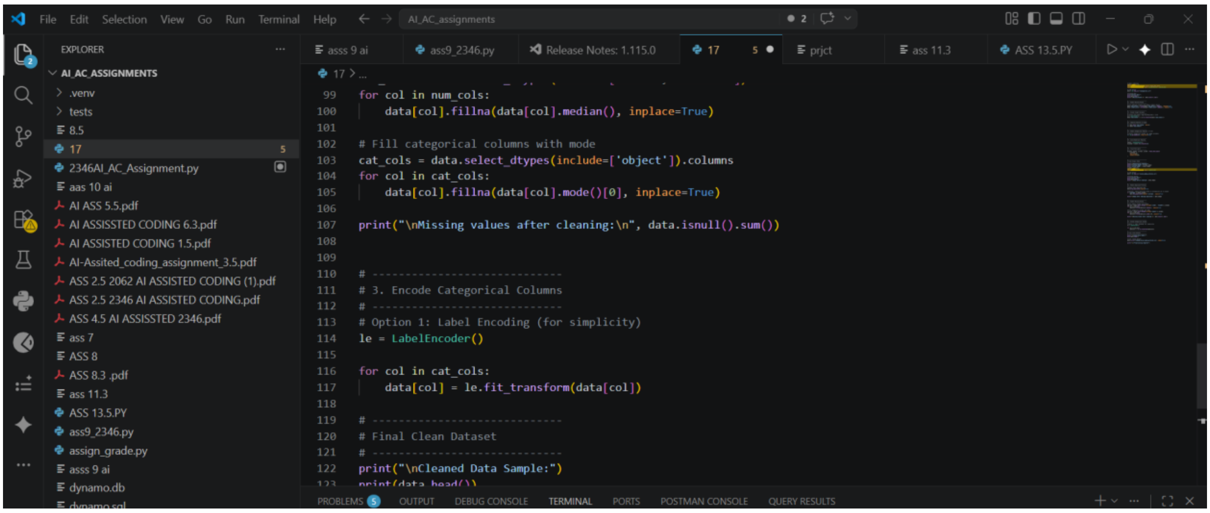
Dataset link:

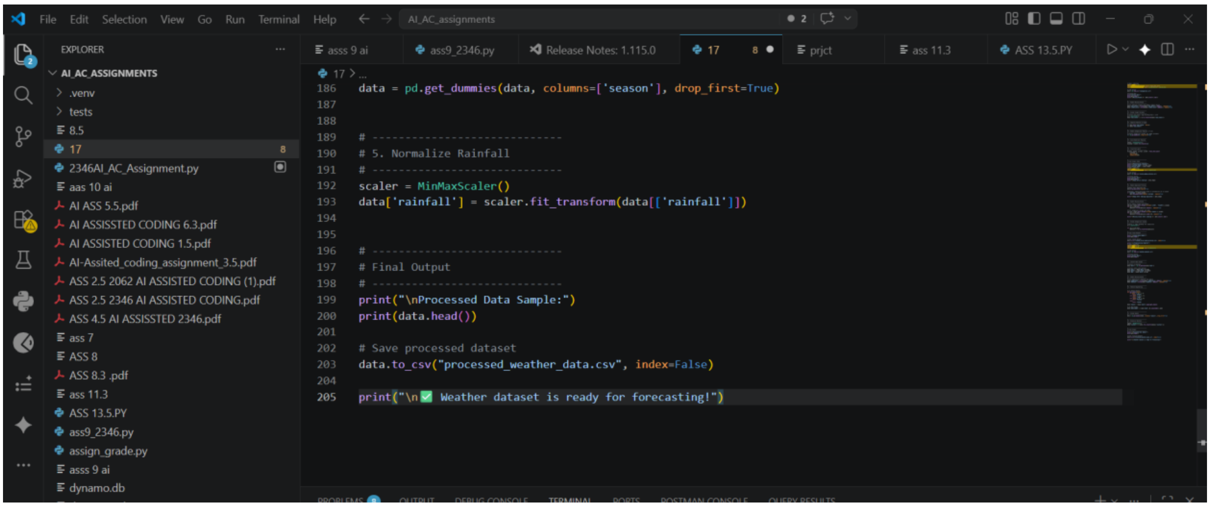
https://www.kaggle.com/datasets/medaxone/top-100-social-

media-profiles

Expected Output:

• A cleaned dataset for engagement prediction.





Movie Ratings Dataset Preparation

Task:

Clean and preprocess a movie ratings dataset using AI-

generated scripts.

Instructions:

• Handle missing values in rating, genre.

• Encode categorical variables like genre, language.

• Remove duplicate movie entries.

• Scale rating values between 0 and 1.

Dataset link:

https://www.kaggle.com/datasets/PromptCloudHQ/imdb-

data

Expected Output:

• A dataset suitable for recommendation systems.